

Effects of Smoking on Outcomes of Hypertension

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Background and Purpose

Hypertension affects nearly half of adults in the United States and is a risk factor for cardiovascular disease and stroke. Smoking can raise its risk by causing damage to blood vessels. We evaluate the association between smoking status and hypertension and assess how this relationship changes after considering confounders such as age, sex, and BMI. This study analyzed cross-sectional data from the 2021-2023 cycles of the National Health and Nutrition Examination Survey.

Methods

Smoking status was categorized as never, former, or current based on questionnaire responses. Logistic regression models were used to estimate odds ratios and 95% confidence intervals for hypertension while adjusting for age, sex, and BMI. Interaction terms assessed whether BMI modified the relationship between smoking and blood pressure.

Results

Our sample size is 6,052 individuals. Current smokers had higher odds of hypertension compared with never-smokers (OR = 1.32, 95% CI: (1.13, 1.54)). Former smoking did not show a significant association with hypertension (OR = 1.06, 95% CI: (0.93, 1.21)). Assessing the interaction term, the effect of BMI on blood pressure is reduced for current smokers compared to non-smokers, suggesting that there exists confounding.

Conclusion

Smoking is associated with increased cardiovascular risk. However, the increase in risk appeared to attenuate among former smokers, as they had a similar risk of hypertension as never-smokers.

Limitations in this study include self-reported smoking status, so longitudinal studies are needed to better assess the temporal relationship between smoking and hypertension. Smokers who have hypertension are encouraged to quit smoking.